NATIONAL TECHNICAL UNIVERSITY OF ATHENS

SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING

LABORATORY OF KNOWLEDGE AND DATABASE SYSTEMS

Academic Year 2025–2026, 6th Semester, ECE

Databases – Semester Assignment (V2)

I. Assignment Description

The General Hospital “Ygeiopolis” has commissioned you to design and implement a system for storing and managing the information required for its operation.

Its staff consists of physicians, nurses, and administrative personnel. The staff members have a social security number (AMKA), first name, last name, age, email, telephone number, hiring date, and personnel type. Physicians additionally have a medical association license number, specialty (e.g., Cardiology, Surgery), and rank (Resident, Consultant B’, Consultant A’, Director). Each physician may have one supervisor, who is also a physician of the hospital. Residents are required to have a supervisor, while Directors do not have supervisors. Cyclical supervision chains are prohibited (e.g., A supervises B, B supervises A).

Each physician may belong to one or more departments. Nurses have ranks (Nursing Assistant, Nurse, Head Nurse) and belong to one department. Finally, the administrative personnel have duties/roles (e.g., Secretary, Accountant), office location, and belong to one department.

The hospital contains departments (e.g., Cardiology, Surgery, ICU, Emergency Department). Each department has a name, description, number of beds, floor/building, and a department director (physician). Each department contains beds with a unique number, type (e.g., ICU, single room, shared room), and status (e.g., available, occupied, under maintenance).

Patients have AMKA, first name, last name, father’s name, age, gender, weight, height, address, telephone number, email, profession, nationality, emergency contact person(s), and insurance provider (e.g., EFKA, private insurance, uninsured). Each patient may have multiple hospitalizations in their medical history.

All hospital departments operate on-call every day (24 hours). Their operation is organized into three eight-hour shifts (morning (07:00–15:00), afternoon (15:00–23:00), and night (23:00–07:00)). For every shift of each department, a specific on-call team must be assigned. Each department shift must be staffed by at least three physicians, six nurses, and two administrative employees.

Additionally, in every shift in which a resident physician participates, at least one physician of rank Consultant A’ or Director must also be present. The system enforces the following maximum monthly shift limits per personnel category: Physicians 15, Nurses 20, and Administrative Personnel 25. Furthermore, between two consecutive shifts of the same staff member there must be a minimum rest period of 8 hours. No staff member may participate in more than 3 consecutive night shifts. The above limits and restrictions are automatically enforced by the system and cannot be bypassed during shift scheduling.

Every patient arriving at the hospital is directed to the Emergency Department and examined by the triage nurse, who records the symptoms and determines the urgency level (1 = immediate, 2 = urgent, 3 = pressing, 4 = less urgent, 5 = non-urgent). Patients are served according to urgency level and, within the same level, in strict order of arrival time (FIFO). The patient either receives instructions and leaves or is referred for hospitalization.

If the patient requires hospitalization, they are admitted to one of the hospital departments and assigned a specific bed. Each hospitalization concerns one patient, one bed, one department, has an admission date, discharge date, admission diagnosis and discharge diagnosis (ICD-10 code and description), as well as total hospitalization cost.

The hospitalization billing follows the Closed Unified Hospitalization system (K.E.N.). Each hospitalization is associated with a KEN code, which determines the base cost and the predicted Average Length of Stay (ALOS). If the patient's actual stay exceeds the ALOS, the system imposes a proportional additional daily charge.

Each hospitalization may be associated with laboratory examinations (e.g., hematological, biochemical, imaging). Each examination has a code, type, date, result (text and/or numerical value with measurement unit), cost, and the physician who ordered it.

Additionally, the hospitalization may be associated with one or more medical procedures/interventions. Each procedure has a code, name, category (surgical, diagnostic, therapeutic), duration, cost, and requires a specific room (operating theater or intervention room). A procedure is performed by one lead surgeon and possibly assistants (physicians or nurses, staff members). Two procedures cannot take place simultaneously in the same room, nor may the same physician participate in two procedures at the same time.

The pharmaceutical treatment that a patient may receive during hospitalization is always administered upon physician instruction (prescribing physician, receiving patient, prescribed medication). The prescription additionally includes dosage, frequency, start date, and end date. Every combination (physician, patient, medication, start date) is unique.

The hospital is required to comply with the database of Article 57 of the European Medicines Agency (EMA), which includes data concerning all medications approved in the European Economic Area (EEA). Patients may have recorded allergies to active substances. Prescribing medication is prohibited if any of its active substances appears in the patient's allergies.

Patients with completed hospitalization may evaluate their experience after discharge. The evaluation is based on a Likert scale at two levels:

(a) Hospitalization evaluation according to the following criteria:

- Quality of nursing care
- Cleanliness
- Food
- Overall experience

(b) Physician evaluation according to the criterion:

- Quality of medical care

The patient may evaluate every physician who prescribed medication during their hospitalization. Evaluation is allowed only for patients with completed hospitalization.

Finally, the hospital is considering creating a corresponding website and requests that all entities be accompanied by corresponding images (e.g., photos of departments, physicians, equipment) with textual descriptions.

II. Implementation Instructions

The implementation specifications may not be fully determined by the hospital. For completeness reasons, you should record any assumptions you make in a table within the README.md file.

You must insert into the Database (DB) information for every entity so that all required queries execute successfully and return appropriate results. If a query does not return data, it will not be graded.

Indicatively required:

- 80 physicians
- 200 patients
- 500 hospitalizations
- 15 departments
- 300 prescriptions
- 10 operating rooms/intervention rooms
- 150 medical procedures
- 200 laboratory examinations

The reference data (e.g., ICD-10, medications, KEN, medical procedures) must be imported exactly as provided from the corresponding sources. Inventing codes or names for these entities is not permitted.

For the allergy component, you may use partial matching (e.g., LIKE in SQL) or simplify the active substances in the data.

For loading the reference files (available in XLS/CSV format), the following is recommended:

- Preprocessing (e.g., with Excel) before import

- Use of LOAD DATA INFILE (MariaDB) or \copy (PostgreSQL) for bulk data import
- For the EMA Article 57 file: pay attention to the separator | in the active substances field, since each active substance must be inserted as a separate record.

III. Requirements

A. Design and implementation of the Database

1. Design the ER diagram resulting from the assignment.

2. Design the relational diagram and develop the database.

2.1.

Define all necessary constraints that ensure the correctness of the database. These include integrity constraints, keys, referential integrity, domain integrity, and user-defined constraints.

2.2.

Define appropriate indexes for the database tables and justify your choices based on their usefulness for the queries in which they are used.

B. Design and execute the following queries

(The queries have equal weight.)

Each query must be implemented with a single query and return one result set. All queries must return valid results; otherwise, they will not be graded.

1. Find the hospital's total revenue per department and per year, analyzed by KEN code (base cost versus additional charge due to exceeding ALOS) and distribution of hospitalizations per insurance provider.

2. For a specific physician specialty, find all physicians belonging to it, indicating whether they had on-call shifts in the current year and how many procedures they performed as lead surgeons.

3. Find which patients have been hospitalized more than 3 times in the same department, together with their total hospitalization cost.

4. For a specific physician, find the average rating from their patients (criterion: quality of medical care) and the overall hospitalization experience.

5. Find the young physicians (age < 35 years) who have performed the highest number of surgical procedures as lead surgeons.

6. For a specific patient, find their hospitalization history, the corresponding diagnoses (ICD-10), the total cost per hospitalization, and their average evaluation score.

7. For each active substance, find the number of patients who have declared an allergy to it and the number of medications containing it, sorted by total number of allergic patients.

8. Find the personnel (physicians, nurses, administrative personnel) who do not have a scheduled on-call shift on a specific date and department.

9. Find which patients were hospitalized for the same number of days within a one-year period, with a total duration exceeding 15 days.

10. Many patients receive combinations of medications. Find the top-3 pairs of active substances that were prescribed simultaneously to the same patient during the same hospitalization, sorted by frequency of occurrence.

11. Find all physicians who have performed at least 5 fewer procedures than the physician with the highest number of procedures in the current year.

12. Find the required number of personnel per department and per shift for a specific week, analyzed by personnel subclass (physicians per specialty, nurses per rank, administrative personnel per role).

13. For every physician, find their complete supervision hierarchy, from the direct supervisor up to the Director, indicating the level at each hierarchy stage.

14. Find which ICD-10 diagnosis categories had the same number of admissions in two consecutive years, with at least 5 cases per year.

15. Find the distribution of triage cases by urgency level, with average waiting time per level, percentage of cases leading to hospitalization, and distribution of referrals per department.

For queries Q4 and Q6, present:

(a) the result of EXPLAIN/EXPLAIN ANALYZE, (b) an alternative version with FORCE INDEX or hint, (c) comparison of estimated cost and actual execution time between the two versions, and (d) a short analysis (3–5 sentences) of your conclusions.

In order for differences in query plans to be visible, you should have a sufficient amount of data.

IV. Procedural Information

- Free selection of groups of up to a MAXIMUM of 3 people (individual work is also allowed) no later than 12/4/2026. Late registrations and/or modifications will not be accepted.

- For the development of your application, it is recommended to use MySQL (MariaDB) or PostgreSQL for database implementation. You may use PHP, Java, Python, or Node.js for implementing parts of the application if you wish.

- Assignments developed using ORM technologies will not be accepted.

- Queries must be implemented elegantly and efficiently in SQL. The use of enums, arrays, json, and xml is not permitted.

- Copying is strictly prohibited. In the event of plagiarism (even partial), all copies of the same assignment will receive a grade of zero. The use of Artificial Intelligence (AI/LLM) tools, such as text generation systems, is permitted only if explicitly declared and properly documented. You must state at the end of the assignment the tool used and the manner in which it contributed to the completion of the assignment. Submission of text or solutions generated by AI/LLM tools without proper acknowledgment is considered a violation of academic ethics rules and is treated the same way as plagiarism.

- Beyond the above requirements, you are free to make assumptions and justify them in order to proceed with your assignment.

V. Deliverables

Assignment submission (until 17/5/2026). Late submissions and/or modifications will not be accepted.

Your submission must contain the following in a zip archive:

1. README.md file (A file containing the documentation of your assignment.)

2. Folder [diagrams]

3. diagrams/er.pdf: E/R Diagram

4. diagrams/relational.pdf: Relational diagram

5. Folder [sql]

6. sql/install.sql: SQL script for creating the database schema

7. sql/load.sql: SQL script for loading the schema with data

8. sql/Qx.sql, sql/Qx_out.txt: For every required SQL query, there must be a Qx.sql file (e.g., Q01.sql) containing your answer, as well as a Qx_out.txt file (e.g., Q01_out.txt) containing the DBMS output.

9. Folder [docs]

10. docs/report.pdf: Document summarizing all steps of your assignment, including screenshots for queries 4 and 6.

11. (Optional) Folder [code] Additional code that may be required for your application to function.

If any file is missing, the corresponding requirements will not be graded.

If you cannot upload the zip file to Helios, you may alternatively submit only a link to the application's Git repository. In this case, the Git repository must follow the above folder and file structure and, if private, you must grant access by the submission deadline.

All submissions are strictly subject to the academic ethics code of NTUA and the School of Electrical and Computer Engineering. Your code must not change from the submission date until grading of the course. If this occurs, your grade will be ZERO (0).

The assignment is provided exclusively within the framework of the teaching semester and may only be submitted during the June examination period. Re-examination/regrading of assignments within the same academic year is not permitted.

VI. Application Demonstration

The application demonstration will take place before the examination period. An exact date will be announced.

For the demonstration:

- The application must be installed and functional on your computer (the use of your own laptop is recommended).
- You must bring a printed report containing at least: E/R diagram, relational diagram, DDL script.
- During the demonstration, you may be asked to answer ad-hoc SQL queries that are not included in the deliverables. The demonstration aims to evaluate understanding of the design choices rather than memorization of solutions.

Attendance at the demonstration is mandatory. No grade will be assigned to any group or student who does not attend.

VII. Grading

The total grade of the assignment is distributed as follows:

1. Up to 90 points for covering all requirements and full application functionality. (Section A: 40 points and Section B: 50 points)

2. Up to 10 points for creativity, application complexity, and excellent presentation.

3. Groups that implement a web application (web UI) for interaction with the database may receive up to 10 additional points.

VIII. Deadlines

All dates are final and no extension will be granted.

No.	Date	Event
1	12/3/2026	Assignment Release
2	12/4/2026	Group Registration
3	17/5/2026	Deliverable Submission
4	18–22/5/2026	Application Demonstration